

"Ridgeline HD: Premature wheel hub bearing wear under heavy load"

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Area: bulletins

1 Condition

Certain Ridgeline HD vehicles operated at or near maximum payload and towing capacity may exhibit premature wear of wheel hub bearing assembly WHB-8580. The customer may report a cyclic humming or growling noise that increases with vehicle speed and changes under cornering load, along with possible play at the affected wheel. Confirm the concern by rotating the wheel and checking for roughness or axial play, then compare against the inspection criteria in Ridgeline HD Suspension Service Manual :: Wheel Hub Bearing Assembly . Review the Ridgeline HD Suspension Service Manual :: Suspension System Overview to rule out related suspension noise sources.

2 Cause

The condition is caused by accelerated raceway and roller fatigue inside wheel hub bearing assembly WHB-8580 when the vehicle is subjected to sustained heavy axle loading beyond typical duty cycles. The increased radial and thrust loads exceed the original bearing's endurance margin, producing the noise and play that characterize WHB-8580 failure. Note that worn suspension components can compound bearing load; if a front clunk or uneven tire wear is also present, review Ridgeline HD: Front clunk and uneven tire wear from worn lower control arm before condemning WHB-8580.

3 Repair Procedure

Replace wheel hub bearing assembly WHB-8580 following procedure PROC-HUB-BEARING and the detailed steps in Ridgeline HD Suspension Service Manual :: Wheel Hub Bearing Replacement Procedure , observing all torque specifications for the axle nut and knuckle fasteners. The applicable service section is described in Ridgeline HD Suspension Service Manual :: Wheel Hub Bearing Sub-System Service . If a strut knock is also noted during inspection, evaluate the strut per Ridgeline HD front strut knock from worn strut assembly . After installing WHB-8580, verify the noise is resolved on a road test and confirm there is no remaining wheel play.